

Behavioral Indicators of Occupational Constraint in Adult Professional English Learning

An Exploratory Longitudinal Corpus Study

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6,045

Total messages

327

Learner-weeks

4

Adult professionals

269–824

Days per learner

RESEARCH PROBLEM

Adult professionals studying English under occupational demands are systematically understudied. Most SLA and learning analytics research focuses on university students in stable institutional settings. Workplace learners face shifting meeting calendars, travel, and deadlines that create irregular participation — a pattern visible in everyday instructional messaging records that has received almost no empirical attention.

CORPUS & METHOD

KakaoTalk instructional message records from four adult professional English learners across automotive, technology, and consumer goods sectors — 6,045 messages, 327 learner-weeks, spanning up to 824 consecutive days of coaching. A manifest-driven R parser produces a message-level corpus with three behavioral indicators:

- ▶ **Schedule-disruption marker** — scheduling-friction language in learner messages
- ▶ **Apology/uncertainty marker** — hedging and apologetic communicative stance
- ▶ **Instructor correction density** — explicitly notated feedback per instructor message

All analyses are within-learner first. Keyword sensitivity analysis tests findings under four keyword definitions (broad, no-late, no-sorry, Korean-only). A parsing log documents raw line counts, continuation merges, and date spans per export.

KEY FINDINGS

- ▶ L002 disruption-apology association is **robust**: SMD +0.75 to +0.81 across all four keyword definitions.
- ▶ Three of four learners show **higher apology/uncertainty language** in disruption-coded weeks, consistent with SRL context-regulation theory.
- ▶ L003 shows a **positive English-proportion trend** over 79 weeks (+0.0026/wk, 95% CI entirely above zero) — the only learner with this pattern.
- ▶ L004 shows **zero Korean-only disruption weeks** across 113 weeks, indicating scheduling friction communicated exclusively in English.
- ▶ Pooled model: apology rate ~ disruption + learner: $\beta = +0.086$ [0.041, 0.132] — exploratory screen only; N=4 precludes population inference.

IMPLICATIONS FOR AI-MEDIATED FEEDBACK

Human coaching interaction in this corpus is scheduling-sensitive, correction-calibrated to individual learner histories, and longitudinally structured. AI feedback tools evaluated against single-session task performance lack access to this temporal dimension. This corpus offers an exploratory benchmark for the interaction structures that AI-mediated language feedback systems might be designed to replicate — and a methodology for collecting such benchmarks at larger scale.

THEORY

Self-Regulated Learning Context regulation under occupational constraint (Pintrich, 2000; Zimmerman, 2000)

Behavioral Trace Analytics Residues of learning in digital interaction records (Clow, 2012)

Participation Theory English-medium participation as learning trace (Sfard, 1998; Wenger, 1998)

FL Anxiety & Feedback Apology stance / corrective interaction (Horwitz et al., 1986; Lyster & Ranta, 1997)

SENSITIVITY

SMD: disruption → apology/uncertainty

	Broad	-late	KO
L001	-0.56	-0.55	-0.54
L002	+0.78	+0.81	+0.81
L003	+0.46	+0.31	+0.38
L004	+0.55	+0.42	NA

L002 stable across all definitions. L004 NA = zero Korean-only disruption weeks.

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PORTFOLIO

- ▶ Reproducible R pipeline
- ▶ Parsing log + validation
- ▶ Sensitivity checks (6 tests)
- ▶ Indicator validation memo
- ▶ Theoretical framing doc
- ▶ Working paper (9 pages)
- ▶ Public GitHub repository

LIMITATIONS

N=4 · Text-only corpus · Proxies not validated · Single instructor · Retrospective design · No outcome measures

NEXT STEPS

N≥15 learners → random-intercept model · Validated disruption + anxiety scales · Pre-registered prospective study · AI feedback system comparison